

Temporal Relevance in Semantic Retrieval: Investigating Implicit Temporal Behaviour in a Standard Dense Retriever

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ABSTRACT

Delivering up-to-date information is a central concern in information retrieval, which is complicated by the varying rates at which information changes: evergreen facts remain valid for years, while time-sensitive facts can become outdated within months. Standard semantic retrievers used in *retrieval-augmented generation* (RAG) systems are designed to find documents with similar meaning, typically neglecting the importance of temporally aligned retrieval. This paper investigates whether such a semantic retriever is inherently sensitive to temporal relevance, and whether this can be induced through temporal cues. The retrieval behaviour is tested across 3 research questions: retrieval without any temporal signal (RQ1), with explicit cues in the queries (RQ2), and finally with timestamps injected into the documents (RQ3). Using a dense retriever over a controlled subset of the ccnews corpus, retrieval recency was compared across 5 query sets with and without temporal cues. Results show no implicit temporal sensitivity and no consistent effect from present-oriented or relative markers. Only absolute year markers and correspondingly injected dates produce a consistent, significant shift toward newer documents. A significant shift was observed primarily where query and document shared an explicit temporal expression, with no comparable shift observed for implicitly conveyed temporal meaning.

1 Introduction

Consider a RAG system asked which football team won this year's Champions League. Because the tournament is decided anew each season, the current answer is vastly outnumbered by topically near-identical documents about past winners. A purely semantic retriever with no explicit temporal signal has little reason to favour the current one over them, feeding the system with wrongful data.

The retriever's error here is not a failure of topical relevance but the inability to recognize that the answer to this question requires recent information. Facts, in general, age at different rates: Evergreen knowledge (i.e. speed of light in a vacuum) is expected to remain valid for many years, if not forever. In contrast, time-sensitive information (i.e. leadership positions, market prices) becomes obsolete in months, days, or even minutes. A retrieval

system indifferent to this distinction fails in both directions: forcing recency onto a stable query surfaces newer but possibly inferior documents [1], while ignoring the age of a document for a dynamic query may produce a confidently wrong context. Even when the correct document is retrieved alongside outdated ones, LLMs relying on a RAG system still face major difficulties when differentiating between current and outdated information [2]. Matching retrieval behaviour to the underlying decay rate of queried information is therefore an essential precondition for reliable retrieval systems.

Although the temporal aspect is often overlooked while focussing on semantically correct retrieval, some research papers propose concepts of time-aware retrieval systems. These systems often either rely on simple reranking after semantic retrieval [3], unnecessarily degrading document quality for evergreen queries, or have a substantial fine-tuning or complexity overhead [4] [5]. As a result, smaller or less resource-intensive RAG systems in practice frequently default to retrievers with no explicit temporal mechanism at all. Whether such a standard, purely semantical retriever nonetheless exhibits any implicit temporal sensitivity remains largely unexamined.

This paper investigates this question along three research questions in which the temporal signal's location shifts at each step. Research question 1 asks, whether semantic retrieval systems exhibit implicit temporal retrieval behaviour, systematically favouring more recent documents for time-sensitive queries compared to evergreen queries, in the absence of any explicit temporal cues. RQ2 then investigates, to what extent temporal cues in queries influence the recency of documents for time-sensitive queries. Lastly, RQ3 asks whether injecting a document's publication date directly into its text can improve retrieval behaviour for time-sensitive queries.

The complete code, as well as dataset, query sets and experiment results are published in a GitLab repository¹.

2. Methodology

2.1 Dataset

The corpus selection was based on two requirements: i) Since this study depends entirely on the state of information in the temporal aspect, each document needs to have an exact timestamp, at which

¹ <https://git.fim.uni-passau.de/motte/data-science-seminar>

its content was up to date. This is distinct from the crawl-date which is included in most available open-web dataset but not sufficient for a meaningful evaluation of the retrieval results. ii) Furthermore, the corpus needed to span multiple years allowing the information to go through a factual drift. *stanford-oval/ccnews*² meets both requirements, providing a large amount of news articles from 2016 through 2024, consistently labelled with the date of publication.

Rather than operating on the whole breadth of the ccnews corpus, the study limits itself to a small English subset of topical domains to keep the experiment controlled. Each domain requires continuous coverage across all years and must include articles about conceptionally stable information as well as evolving facts. Taking this into account, the following three topics were chosen: financial markets, internet platforms and UK football.

Articles without affiliation to one of those topics were filtered out of the dataset. To track down relevant documents, the titles were matched against domain-specific keyword lists with their own constraints: granular enough for multiple articles to be relevant to a query, allowing the retriever to choose between a variety of candidates of differing ages, yet broad enough to find enough matches. The keywords themselves could not cluster in time. The representation of a term like “covid-19” would be heavily unbalanced, therefore violating the goal of an even distribution across all years.

Finally, to prevent the corpus itself from introducing a quantitative temporal artefact, articles were limited to exactly 3700 documents per topic and per year. Without this constraint, retrieval could appear recency-biased simply because more articles were crawled in recent years than for old ones. After excluding the year 2016 for insufficient coverage of the selected topics, a final dataset of $3700 \text{ articles} \times 3 \text{ topics} \times 8 \text{ years} \approx 90k \text{ documents}$ was obtained.

2.2 Query construction

Multiple query sets are constructed that serve as the basis for the retrieval experiments. All queries are created based on the same keyword lists used for domain filtering and reviewed manually.

First, an **evergreen** set serves as the control group against the time-sensitive sets by querying only stable facts.

The foundation queries for the other sets are made up of **time-sensitive** questions without any cues pointing to the need of recent documents. Their temporal dimension is therefore solely implied by the evolving nature of the queried facts whose answers change over time, testing the retriever’s implicit sense of time. In this paper, time-sensitiveness is confined to the aspect of recency rather than historical alignment. The two categories are defined in respect to the timespan covered by the dataset: evergreen answers are expected to remain stable across a minimum of 8 years (i.e. photosynthesis), whereas time-sensitive ones will plausibly change within that time window or in fixed intervals (i.e. presidency).

The time-sensitive set is then used as a basis for 3 additional sets by extending every question with each of three different types of temporal cues. Each marker type constitutes its own query set:

- **present-oriented** (i.e. “currently”, “latest”),
- **relative** (i.e. “since last year”) and
- **absolute** (“as of 2024”, the cutoff date of the dataset).

For each of the three topic domains, we construct 20 questions per query type, yielding 60 evergreen, 60 unmarked time-sensitive queries and 60 queries each for the three types of temporal cues (300 queries total).

2.3 Date Injection

RQ1 and RQ2 test the differences of temporal retrieval behaviour at query-level, while RQ3 focuses on the documents themselves, testing whether the retriever can improve its understanding of temporal intent when an explicit signal is embedded into the text.

The signal is introduced by inserting the sentence “Published on {publication date}.” between the article’s title and body prior to embedding. The individual publication date is written as “Month dd, yyyy” (i.e. “August 12, 2022”), representing a realistic style for news articles.

For an article that takes up the remaining 512-token budget, the single injection accounts for approximately 1.4-1.6% of the total token count.

2.4 Retrieval

With all the data prepared, retrieval is performed using *sentence-transformers/multi-qa-mpnet-base-dot-v1*³, a model trained for the embedding of short queries and longer passages. This model truncates documents longer than 512 tokens, however the most important information in news articles is mostly at the beginning allowing the results to still be representative [6]. Document embeddings are L2-normalized and indexed with *FAISS*’s *IndexFlatIP*, an exact inner-product index equivalent to cosine similarity for normalized vectors.

To preserve the isolation of topics established during corpus construction and eliminating cross-topic interference, a separate index is built for each of the three topic-domains rather than merging them into one.

During retrieval, each element in the selected query set is encoded with the same model and $k \in \{1, 10, 20, 50, 100\}$ documents with the highest similarity score are retrieved. By running every experiment at multiple retrieval depths and capturing the results separately, the consistency of any observed effect can be examined.

2.5 Evaluation

The concept of the evaluation is identical across all three research questions: whether the ages of the retrieved documents differ systematically between two conditions, intuitively investigating if

² <https://huggingface.co/datasets/stanford-oval/ccnews>

³ <https://huggingface.co/sentence-transformers/multi-qa-mpnet-base-dot-v1>

the retriever adjusts its behaviour to fit the temporal intent of the user. This is formalized with the following pair of hypotheses, which is instantiated separately across all experiments listed in Table 1:

H₀: There is no stochastic difference between the weighted average age of documents retrieved under the comparison condition and the baseline condition.

H₁: Such a difference exists.

As a metric for the evaluation, the *score-weighted average publication age* over the top-k retrieved documents is computed for each individual query, weighting each article’s contribution by its retrieval relevance:

$$\bar{a}_{weighted} = \sum_{i=1}^k w_i \cdot a_i, \quad w_i = \frac{s_i}{\sum_{j=1}^k s_j},$$

where a_i is defined as the number of days between the publication date of the i -th retrieved document and a fixed date outside the corpus’ cutoff date (01.01.2025). s_i denotes its corresponding retrieval score, and k the top-k value used in this experiment.

For RQ1, the $\bar{a}_{weighted}$ values of unmarked time-sensitive queries are compared against the values for the evergreen queries using a two-sided Mann-Whitney U test and the rank-biserial correlation for unpaired values.

The results of the experiments for RQ2 and RQ3 are evaluated with the Wilcoxon signed-rank test and rank-biserial correlation for paired values.

Across all experiments, the significance threshold for the statistical tests is set as $\alpha = 0.05$. The rank-biserial correlation serves as an indicator for effect size and is reported as r_{rb} . A positive effect size means that the comparison group tends to have smaller $\bar{a}_{weighted}$ values than the baseline, indicating a trend towards more recent documents for the comparison group.

Research Question	Comparison	Baseline	Fixed
RQ1	TS, no cues	Evergreen	Non-injected index
RQ2	TS, present-oriented	TS, no cues	Non-injected index
	TS, relative	TS, no cues	Non-injected index
	TS, absolute	TS, no cues	Non-injected index
RQ3	Injected index	Non-injected index	Evergreen
	Injected index	Non-injected index	TS, no cues
	Injected index	Non-injected index	TS, absolute

Table 1: Overview of experiment conditions

3 Results

3.1 RQ1 Results

Across all five evaluated top-k values, the Mann-Whitney U tests revealed no statistically significant difference in the temporal distribution of retrieved documents between evergreen and time-sensitive queries without any temporal cues (Table 2). At the topic level, Financial Markets showed no consistent trend, whereas Internet Platforms exhibited a non-significant trend toward more recent documents for time-sensitive queries, and UK Football toward older documents. These opposing tendencies likely account for the near-null pooled effect.

		top-k				
		1	10	20	50	100
pooled	r_{rb}	0.14	0.01	0.03	0.00	0.02
	p	0.20	0.90	0.80	0.99	0.85
financial markets	r_{rb}	0.20	0.06	-0.02	-0.05	-0.06
	p	0.29	0.76	0.95	0.80	0.76
internet platforms	r_{rb}	0.19	0.11	0.19	0.26	0.33
	p	0.32	0.56	0.32	0.16	0.08
UK football	r_{rb}	0.00	-0.20	-0.17	-0.23	-0.24
	p	1.00	0.29	0.38	0.22	0.19

Table 2: Retrieval results of TS without cues versus Evergreen (non-injected index)

3.2 RQ2 Results

For RQ2, the three temporal-marker conditions applied to time-sensitive queries are compared against their unmarked counterparts using paired Wilcoxon signed-rank tests (Table 3). Present-oriented cues yield no consistent shift in retrieval recency: the only significant result can be observed at $k = 50$ ($r_{rb} = 0.41$, $p = 0.006$), where UK Football is the only domain reaching significance individually. Relative cues show no significant pooled effect at any depth, with small and inconsistent effect sizes. Absolute cues produce the clearest pattern, retrieving significantly newer documents than the baseline at every tested depth, peaking at $k = 50$. This effect is not uniform across domains, however: it is concentrated in Financial Markets and UK Football, both significant from $k = 10$ onward, while Internet Platforms shows the same positive shift without significance at any depth.

		top-k				
		1	10	20	50	100
present-oriented	r_{rb}	0.50	0.13	0.09	0.41	0.20
	p	0.14	0.37	0.54	0.006	0.18
relative	r_{rb}	0.38	-0.17	-0.25	-0.01	0.06
	p	0.11	0.25	0.09	0.97	0.68
absolute	r_{rb}	0.46	0.80	0.81	0.85	0.74
	p	0.02	<0.001	<0.001	<0.001	<0.001

Table 3: Topic-pooled retrieval results of TS with three types of temporal cues versus TS without cues (non-injected index)

3.3 RQ3 Results

Table 4 reports the paired Wilcoxon signed-rank comparison between the injected and non-injected indices across three query categories.

For evergreen queries, no significant difference appears up to $k = 20$. However, at $k = 50$ and $k = 100$, the injected index returns a significantly older set of top-k documents. Time-sensitive queries without temporal cues show no significant shift at any cutoff, with effect sizes remaining close to zero throughout. Queries carrying explicit absolute cues show the clearest pattern: the difference is not significant at $k = 1$, but from $k = 10$ onward the injected index consistently returns newer documents, with the effect strengthening from $r_{rb} = 0.66$ at $k = 10$ to $r_{rb} = 0.99$ at $k = 100$.

This pattern holds within each of the three topic domains individually, not only in the pooled results.

		top- <i>k</i>				
		1	10	20	50	100
Evergreen	r_{rb}	-0.21	0.04	-0.18	-0.49	-0.51
	p	0.53	0.80	0.23	<0.001	<0.001
TS (no cues)	r_{rb}	0.03	-0.05	-0.06	0.08	-0.17
	p	0.91	0.72	0.66	0.61	0.24
TS (absolute cues)	r_{rb}	0.47	0.66	0.74	0.96	0.99
	p	0.11	<0.001	<0.001	<0.001	<0.001

Table 4: Topic-pooled retrieval results for injected versus non-injected index (for Evergreen, TS without cues, TS with absolute cues)

Further investigation of the results for absolute time-sensitive queries revealed that the documents retrieved from the injected index are not concentrated around the injected year but are redistributed across the full range of publication dates. Relative to the non-injected index, retrieval mass for old documents is considerably smaller. This gap is closed gradually for documents from 2019 to 2021, with results of the injected index eventually rising above the baseline’s retrieval mass and peaking at the year 2024 (Figure 1).

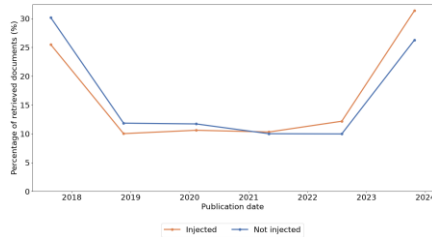


Figure 1: Publication-date distribution of retrieved documents for absolute TS queries on the injected and non-injected index (binned)

4. Discussion

The three experiments differ in where the temporal signal is placed: implicit in the query’s topic (RQ1), in its wording (RQ2), or embedded in the document text (RQ3). Across all three, the same pattern emerges: retrieval shifts toward newer documents only where the query contains absolute cues, not where temporal meaning is present without one. There was no evidence of a difference in retrieval behaviour for time-sensitive topics alone as well as relative cues. Similarly, injected publication dates paired with unmarked queries resulted in a null finding. The only significant result for queries with present-oriented (i.e. “current”) cues was an isolated exception. Only absolute cues, which name an explicit year, produced a consistent shift towards recency. This trend towards newer documents was especially strong once every document in the index was injected with its publication date.

The null hypotheses could therefore not be rejected for time-sensitive queries wherever the temporal signal remained implicit. Only with absolute cues could H_0 be rejected for most retrieval depths at a significance level of $\alpha = 0.05$.

At first, the drastic results for the absolute query set could be explained by the retriever matching the year as an ordinary lexical token rather than accounting for its temporal meaning. However, a detailed visualization of the distribution of publication dates for the non-injected versus injected index contradicted this explanation: It showed a uniform trend towards newer documents instead of an isolated increase around the injected year. One possible explanation is that the inserted dates introduce a graded temporal signal into the document’s embeddings.

A less expected result is that date injection shifted evergreen retrieval toward older documents at greater depth. The underlying cause of this effect could not be determined and could be investigated further in future work.

5. Limitations

Several limitations qualify these findings. First, the average age of retrieved documents metric used for testing is only a proxy for temporal relevance. Newer documents are more likely to reflect the current state of a fact, but older documents can also remain accurate if the given fact hasn’t changed since publication. An observed age difference can therefore only be interpreted as an indication of temporally appropriate retrieval behaviour, not as a direct proof.

Second, the distribution of publication dates in the corpus itself is not even. Although articles were balanced by year, their publication dates cluster around a small number of crawling dates. This is already observable in the ccnews dataset and reflects the collection strategy used by stanford-oval.

A third limitation is that queries were manually written and assigned to the evergreen and time-sensitive categories, which introduces classification noise and subjective judgement into the experiments. Furthermore, the two categories naturally exist as two ends of a continuum rather than a binary classification.

Finally, the RQ3 date-injection procedure is constrained by the embedding model’s limit of 512 tokens. Longer articles may be truncated after inserting the date tag in the front, resulting in lost content compared to the original articles. Additional indexes with matching paddings to apply the same truncation to the baseline documents was not computationally feasible.

6. Deviation from Initial Research Question

The initial formulation of RQ3 proposed evaluating recency-based reranking within a RAG pipeline against a manually curated ground truth of time-sensitive queries. However, this design proved infeasible for two related reasons. First, the corpus’s temporal density decays sharply toward the cutoff date, leaving only a small number of documents that reflect the current state of any given fact. This sharply limits the pool of time-sensitive queries for which a valid ground truth document could be constructed in the first place. Second, even after adapting the criteria toward slower-drifting facts to reduce this dependency, a further issue emerged: given the corpus’s limited size, a document selected as ground truth for a query is, in practice, very likely to already be among the most semantically similar documents to it, leaving little headroom for a reranking step to produce a measurable effect.

The concern that any produced effect would therefore have no evidentiary value caused the final research question to be redirected into another direction.

7. Conclusion

This paper examined whether a standard, purely semantic retriever implicitly adapts to the different decay rates of evergreen and time-sensitive information, and whether such behaviour can be induced through simple temporal signals. RQ1 found no evidence of implicit temporal sensitivity: retrieval for time-sensitive queries was not systematically more recent than for evergreen ones. The second research question showed that this insensitivity mostly persists when time-sensitive queries carry temporal cues. Only absolute markers produced a consistent shift toward newer documents. RQ3 confirmed this pattern from the document side: injecting an explicit publication date shifted retrieval only for queries that themselves named a year, leaving unmarked time-sensitive queries unaffected. Across all three research questions, the retriever proved responsive only to literal, shared temporal expressions, never to temporal meaning conveyed implicitly through topic or wording alone.

AI Disclaimer

Generative AI was used to assist with language editing and structuring, methodological guidance, query construction and code development for the experiments.

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